

The background is a blurred image of a computer monitor. It shows a dark-themed code editor with some text in green and blue. At the bottom, a Windows taskbar is visible with various application icons like Chrome, Photoshop, and others.

Intoduction To Programming Using C++ Level 2 Delete Record From File



Introduction To Programming Using C++

Level 2

Delete Record From File

Definition

 `DeleteRecordFromFile()`

 `DeleteRecordFromFile()` is a function used to delete a specific record from a text file.

 The function loads the file contents into a vector, finds the record that should be deleted, removes it from the data, and then saves the remaining data back to the file.

Explanation

Modifying File Data

 In these lessons, we do not directly remove a line from the middle of a text file.

 Instead, we load the file contents into a `vector<string>`, modify the vector in memory, and then rewrite the file.

`PrintFileContent()`

 Reads and displays the file contents.

 It is mainly used here to show the contents before and after deletion.

`LoadDataFromFileToVector()`

 Reads the file line by line using `getline()`.

 Each line is stored as an element inside the vector.

`SaveVectorToFile()`

 Opens the file using `ios::out`.

 Writes the remaining vector elements back to the file.

 Existing file contents are replaced by the new vector contents.

`DeleteRecordFromFile()`

 Searches for the requested record inside the vector.

 When the record is found, its value is changed to an empty string.

 `SaveVectorToFile()` skips empty strings when rewriting the file.

Procedures

1. Load the file contents into a `vector<string>`.
2. Loop through all vector elements.
3. Compare each element with the record we want to delete.
4. If a match is found, set that element to an empty string.
5. Save the modified vector back to the file.
6. The empty element is skipped when the file is rewritten.

💡 Example

```
void DeleteRecordFromFile(string FileName, string Record)
{
    vector<string> vFileContent;
    LoadDataFromFileToVector(FileName, vFileContent);
    for (string& Line : vFileContent)
    {
        if (Line == Record)
        {
            Line = "";
        }
    }
    SaveVectorToFile(FileName, vFileContent);
}

int main()
{
    cout << "File Content Before Delete:\n";
    PrintFileContent("MyFile.txt");
    DeleteRecordFromFile("MyFile.txt", "Ali");
    cout << "\n\nFile Content After Delete:\n";
    PrintFileContent("MyFile.txt");
    return 0;
}
```

💡 Example

➡ Result

Before deletion:

Ali

Shadi

Maher

Fadi

Lama

After deleting `Ali` :

Shadi

Maher

Fadi

Lama

Note

- ◆ The record is not physically removed directly from the file; the vector is modified first.
- ◆ ``Line = ""`` marks the record as empty.
- ◆ ``SaveVectorToFile()`` ignores empty strings when writing the vector back to the file.
- ◆ This technique is useful for simple text-file record management.

⊘ Warning

- ◆ If the record does not exist, no element is changed.
- ◆ If the same record appears more than once, this function will remove all matching records.
- ◆ The file should be successfully loaded before attempting to modify and save its contents.



Comparison

▨ Delete Record

- ✓ Finds the requested record.
- ✓ Changes the matching element to `""`.
- ✓ Empty elements are skipped when saving.
- ✓ The record disappears from the rewritten file.

▤ Update Record

- ✓ Finds the requested record.
- ✓ Replaces the matching element with a new value.
- ✓ The updated element is written back to the file.
- ✓ The record remains in the file with a different value.

✦ Summary

- ✓ Load the file into a vector.
- ✓ Search for the record.
- ✓ Set the matching element to an empty string.
- ✓ Save the modified vector back to the file.
- ✓ `SaveVectorToFile()` skips empty elements.
- ✓ The record is therefore removed from the resulting file.

Thank you 😊